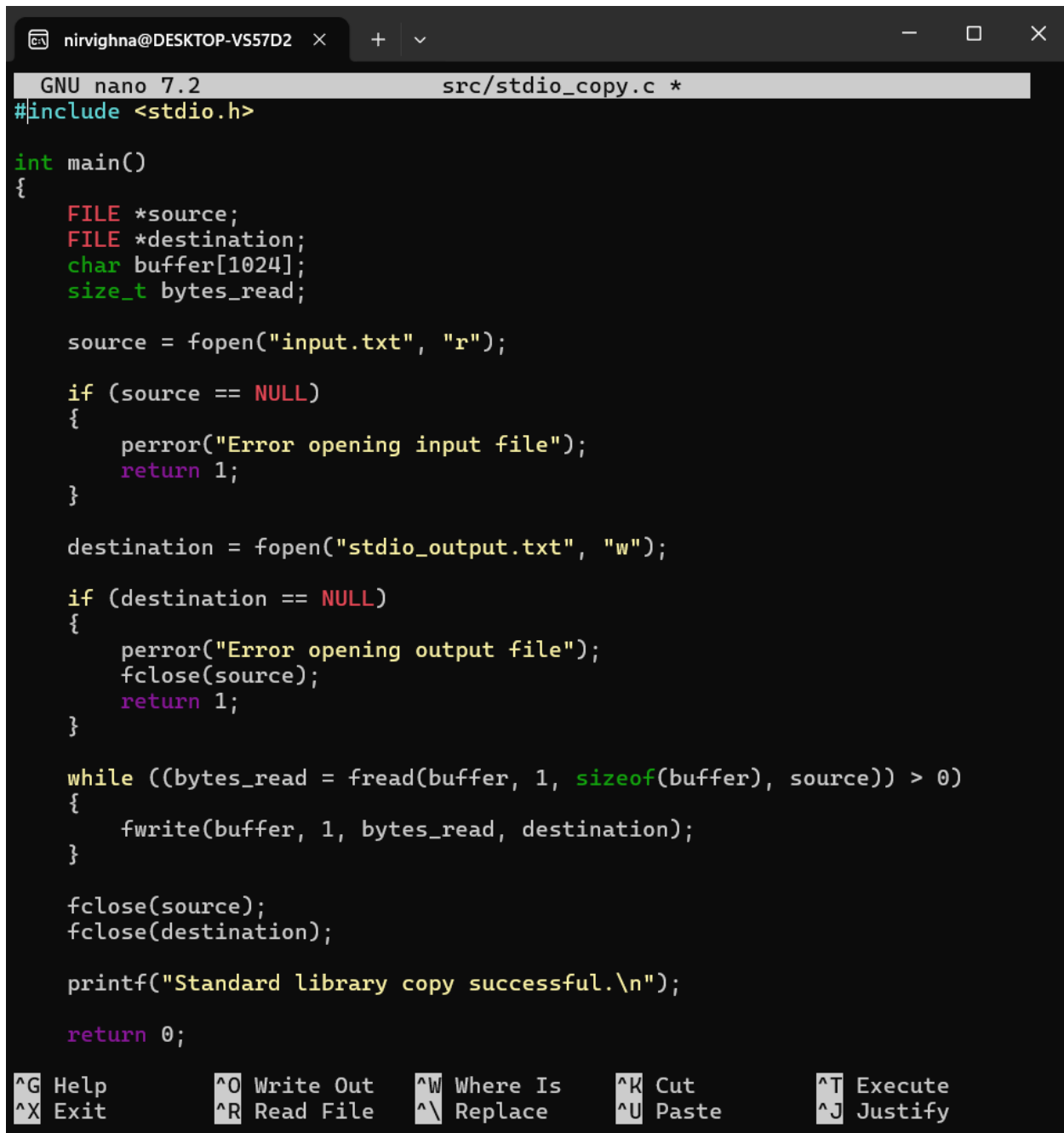


PRACTICAL 9:

```
nirvighna@DESKTOP-VS57D2I:~$ cd ~/OS-Week1
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l src/file_copy.c bin/file_copy
-rwxr-xr-x 1 nirvighna nirvighna 18080 Sep 25 17:22 bin/file_copy
-rw-r--r-- 1 nirvighna nirvighna 1328 Sep 25 17:21 src/file_copy.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ echo "This is Practical Session 9 using low-level Linux file I/O." > input.txt
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat input.txt
This is Practical Session 9 using low-level Linux file I/O.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/file_copy
File copied successfully.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat output.txt
This is Practical Session 9 using low-level Linux file I/O.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cmp input.txt output.txt
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```



```
GNU nano 7.2 src/stdio_copy.c *
#include <stdio.h>

int main()
{
    FILE *source;
    FILE *destination;
    char buffer[1024];
    size_t bytes_read;

    source = fopen("input.txt", "r");

    if (source == NULL)
    {
        perror("Error opening input file");
        return 1;
    }

    destination = fopen("stdio_output.txt", "w");

    if (destination == NULL)
    {
        perror("Error opening output file");
        fclose(source);
        return 1;
    }

    while ((bytes_read = fread(buffer, 1, sizeof(buffer), source)) > 0)
    {
        fwrite(buffer, 1, bytes_read, destination);
    }

    fclose(source);
    fclose(destination);

    printf("Standard library copy successful.\n");

    return 0;
}
```

 ^G Help	 ^O Write Out	 ^W Where Is	 ^K Cut	 ^T Execute
 ^X Exit	 ^R Read File	 ^\ Replace	 ^U Paste	 ^J Justify

```
nirvighna@DESKTOP-VS57D2  ×  +  ▾
GNU nano 7.2 src/redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int fd;

    fd = open("redirect_output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    if (dup2(fd, STDOUT_FILENO) == -1)
    {
        perror("dup2");
        close(fd);
        return 1;
    }

    close(fd);

    printf("This output has been redirected to a file.\n");
    printf("dup2() successfully redirected standard output.\n");

    return 0;
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/redirect
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat redirect_output.txt
This output has been redirected to a file.
dup2() successfully redirected standard output.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat redirect_output.txt
This output has been redirected to a file.
dup2() successfully redirected standard output.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin/file_copy bin/stdio_copy bin
/redirect
-rwxr-xr-x 1 nirvighna nirvighna 18080 Sep 25 17:22 bin/file_copy
-rwxr-xr-x 1 nirvighna nirvighna 17568 Sep 29 18:45 bin/redirect
-rwxr-xr-x 1 nirvighna nirvighna 19000 Sep 29 18:39 bin/stdio_copy
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```